

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	K.Karthik	Reg. No.:	192525040
Section / Batch:	AI & ML	Date:	24/07/2026

Code of Conduct

I K.Karthik, 192525040 (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = 0;
7
8 int pop() {
9     if (top == 0) {
10         printf("Underflow\n");
11         return -1;
12     }
13     return stack[--top];
14 }
15
16
17 int main() {
18     stack[0] = 10;
19     stack[1] = 20;
20     stack[2] = 30;
21     top = 3;
22
23     printf("Popped element: %d\n", pop());
24     printf("Popped element: %d\n", pop());
25 }
```

Stack: 10 20 30

Output

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Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = 0;
7
8 int pop() {
9     if (top == 0) {
10         printf("Underflow\n");
11         return -1;
12     }
13
14     return stack[--top];
15 }
16
17 int main() {
18     stack[0] = 10;
19     stack[1] = 20;
20     stack[2] = 30;
21     top = 3;
22
23     printf("Popped element: %d\n", pop());
24     printf("Popped element: %d\n", pop());
25 }
```

Stack: 10 20 30

Output

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Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init() {
7     top = 0;
8 }
9
10 int main() {
11     init();
12
13     printf("Stack initialized successfully.\n");
14     printf("Top = %d\n", top);
15
16     return 0;
17 }
```

Submitted

Test Input

No input required

Output

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Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

```
1 #include <stdio.h>
2 #include <ctype.h>
3
4 #define MAX 100
5
6 char stack[MAX];
7 int top = -1;
8
9 void push(char ch) {
10     stack[++top] = ch;
11 }
12
13 char pop() {
14     if (top == -1)
15         return '\0';
16     return stack[top--];
17 }
18
19 int main() {
20     char infix[MAX], postfix[MAX];
21     int i, j = 0;
22     char ch;
23
24     printf("Enter infix expression: ");
```

(A+B)*C

Output

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Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

```
1 #include <stdio.h>
2
3 int main() {
4     char s[20];
5     char stack[20];
6     int top = -1;
7     int i, ok = 1;
8
9     printf("Enter brackets: ");
10    scanf("%19s", s);
11
12    for (i = 0; s[i] != '\0'; i++) {
13        if (s[i] == '(' || s[i] == '[' || s[i] == '{') {
14            stack[++top] = s[i];
15        }
16        else if (s[i] == ')' || s[i] == ']' || s[i] == '}') {
17            if (top == -1) {
18                ok = 0;
19                break;
20            }
21
22            if ((s[i] == ')') && stack[top] == '(') ||
23                (s[i] == ']') && stack[top] == '[') ||
24                (s[i] == '}') && stack[top] == '{') {
25                top--;
```

Enter brackets: {[()]}

Output

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Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

1#include <stdio.h>

2#include <stdlib.h>

3

4struct Node {

5 int data;

6 struct Node *next;

7};

8

9int main() {

10 struct Node *head, *second, *third, *temp;

11

12 head = malloc(sizeof(struct Node));

13 second = malloc(sizeof(struct Node));

14 third = malloc(sizeof(struct Node));

15

16 head->data = 10;

17 head->next = second;

18

19 second->data = 20;

20 second->next = third;

21

22 third->data = 30;

23 third->next = NULL;

24

10 20 30

Output

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Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head, *second, *third, *temp;
11
12     head = malloc(sizeof(struct Node));
13     second = malloc(sizeof(struct Node));
14     third = malloc(sizeof(struct Node));
15
16     head->data = 10;
17     head->next = second;
18
19     second->data = 20;
20     second->next = third;
21
22     third->data = 30;
23     third->next = NULL;
24
25 }
```

10 20 30

Output

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5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

your fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10    }
11
12    return stack[top--];
13 }
14
15 int main() {
16     int x = pop();
17
18     if (x != -1)
19         printf("Popped element = %d", x);
20
21     return 0;
22 }
```

test input

Stack: 10 20 30
top = 2

Output

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Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10    struct Node *head, *second, *third;
11    struct Node *current, *prev, *next;
12
13    head = malloc(sizeof(struct Node));
14    second = malloc(sizeof(struct Node));
15    third = malloc(sizeof(struct Node));
16
17    head->data = 10;
18    head->next = second;
19
20    second->data = 20;
21    second->next = third;
22
23    third->data = 30;
24    third->next = NULL;
```

10 20 30

Output

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5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

```
1 #include <stdio.h>
2
3 int queue[5] = {10, 20, 30};
4 int front = 0;
5 int rear = 2;
6
7 int dequeue() {
8     if (front == -1) {
9         printf("Underflow");
10        return -1;
11    }
12
13    return queue[front++];
14 }
15
16 int main() {
17     int x = dequeue();
18
19     if (x != -1)
20         printf("Deleted element = %d", x);
21
22     return 0;
23 }
```

Queue: 10 20 30
front = 0

Output

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Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int peek() {
7     if (top == -1) {
8         printf("Stack Underflow");
9         return -1;
10    }
11    return stack[top];
12 }
13
14 int main() {
15     printf("Top element = %d", peek());
16     return 0;
17 }
```

Submitted

Test Input

Stack: 10 20 30
top = 2

Output

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Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = 0;
7 int rear = 2;
8
9 int main() {
10     int vehicle = 40;
11
12     if ((rear + 1) % MAX == front) {
13         printf("Queue Full");
14     } else {
15         rear = (rear + 1) % MAX;
16         queue[rear] = vehicle;
17         printf("Vehicle inserted = %d", vehicle);
18     }
19
20     return 0;
21 }
```

Submitted

Test Input

```
front = 0
rear = 2
vehicle = 40
```

Output

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5 marks

Q4

Stack

5 marks

5 marks

Q5

Doubly Linked List

5 marks

5 marks

Q6

Queue

5 marks

5 marks

Q7

Queue - Deque

5 marks

5 marks

Q8

Binary Search

5 marks

5 marks

Q9

Circular Queue

5 marks

Submitted

your fixed code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow");
9         return -1;
10    }
11
12    return stack[top--];
13 }
14
15 int main() {
16     int x = pop();
17
18     if (x != -1)
19         printf("Popped element = %d", x);
20
21     return 0;
22 }
```

test input

Stack: 10 20 30
top = 2

Output

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⌵

Q5

Doubly Linked List

5 marks

✓

Q6

Queue

5 marks

✓

Q7

Queue - Deque

5 marks

✓

Q8

Binary Search

5 marks

✓

Q9

Circular Queue

5 marks

✓

Q10

Stack - Postfix Expression

5 marks

✓

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *prev;
7     struct Node *next;
8 };
9
10 void deleteNode(struct Node **head, struct Node *del) {
11     if (del == NULL)
12         return;
13
14     if (*head == del)
15         *head = del->next;
16
17     if (del->prev != NULL)
18         del->prev->next = del->next;
19
20     if (del->next != NULL)
21         del->next->prev = del->prev;
22
23     free(del);
24 }
```

List: 10 <-> 20

Delete: 20

Output

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Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

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```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int rear = 0;
7
8 void enqueue(int data) {
9     if (rear == MAX) {
10         printf("Queue Overflow");
11         return;
12     }
13
14     queue[rear++] = data;
15 }
16
17 int main() {
18     enqueue(10);
19     enqueue(20);
20
21     printf("Queue: %d %d", queue[0], queue[1]);
22
23     return 0;
24 }
```

10 20

Output

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Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int deque[MAX];
6 int front = 2;
7 int rear = 3;
8
9 void insertFront(int data) {
10     if ((front == 0 && rear == MAX - 1) ||
11         (front == rear + 1)) {
12         printf("Deque Full");
13         return;
14     }
15
16     if (front == 0)
17         front = MAX - 1;
18     else
19         front = front - 1;
20
21     deque[front] = data;
22 }
23
24 int main() {
```

Data = 10

Output

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Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x) {
4     if (l > r)
5         return -1;
6
7     int mid = l + (r - l) / 2;
8
9     if (arr[mid] == x)
10        return mid;
11
12    if (arr[mid] > x)
13        return binarySearch(arr, l, mid - 1, x);
14
15    return binarySearch(arr, mid + 1, r, x);
16 }
17
18 int main() {
19     int arr[] = {10, 20, 30, 40, 50};
20     int x = 40;
21     int result;
22
23     result = binarySearch(arr, 0, 4, x);
24 }
```

Array: 10 20 30 40 50
Search: 40

Output

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Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX] = {10, 20, 30};
6 int front = 0;
7 int rear = 2;
8
9 int dequeue() {
10     if (front == -1)
11         return -1;
12
13     int data = queue[front];
14
15     if (front == rear) {
16         front = rear = -1;
17     } else {
18         front = (front + 1) % MAX;
19     }
20
21     return data;
22 }
23
24 int main() {
```

Queue: 10 20 30
front = 0
rear = 2

Output

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Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

Your Fixed Code

```
1 #include <stdio.h>
2
3 int main() {
4     char exp[20];
5     int stack[20], top = -1;
6     int i, a, b, result;
7
8     printf("Enter postfix: ");
9     scanf("%19s", exp);
10
11     for (i = 0; exp[i] != '\0'; i++) {
12         if (exp[i] >= '0' && exp[i] <= '9') {
13             stack[++top] = exp[i] - '0';
14         } else {
15             b = stack[top--];
16             a = stack[top--];
17
18             if (exp[i] == '+')
19                 result = a + b;
20             else if (exp[i] == '-')
21                 result = a - b;
22             else if (exp[i] == '*')
23                 result = a * b;
24             else if (exp[i] == '/')
25                 result = a / b;
```

Test Input

23+

Output

Enter postfix: Result = 5

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